University of Belgrade, Faculty of Sport and Physical Education

In-class work — week 1

Statistics in Physical Education and Sport · chapters 2–5 · jamovi, about 25 minutes

The data are textbook examples (Weir & Vincent, 2021), in the files STAT_DAS_N01_Class_*.xlsx (one file per data set). For each step, write down the number or sentence asked for; we check them at the end of the class.

Pull-ups, N = 212 (file STAT_DAS_N01_Class_Pullups.xlsx; table 2.2)

1. Set the type of variable X to Ordinal; Exploration → Descriptives → Frequency tables.
2. Write down: how many students did 10 pull-ups? Which score has the highest frequency (the mode)?
3. Plots → Bar plot gives one bar per value; picture a line through the tops of the bars — that is the frequency polygon. In one sentence: is the distribution roughly symmetrical?

100-m swim, N = 79 (file STAT_DAS_N01_Class_Swim.xlsx, seconds to the nearest tenth; table 2.4)

4. Select X, Data → Transform: new variable Interval, Create New Transform, formula FLOOR(($source + 0.5) / 5) * 5 (lower apparent limit of an interval of size 5, by real limits: 80–84 runs from 79.5 to 84.49…), Measure type Ordinal. (Compute cannot create an ordinal variable.)
5. Frequency tables for Interval: write down the frequency of the interval 85–89. Which interval does 84.7 s go into?
6. Plots → Histogram for X: compare its shape with the frequency table.

Free throws, N = 60 (file STAT_DAS_N01_Class_Free_throws.xlsx; table 3.2)

7. Frequency tables: the percentile rank of a score of 7 is in the Cumulative % column. Write it down.
8. Which score is the 75th percentile (the first row where Cumulative % ≥ 75)?
9. Descriptives: Mode, Median, Mean, Std. deviation. Write down all four numbers and, in one sentence, why the mean and the median differ (or do not).

References

Weir, J. P., & Vincent, W. J. (2021). *Statistics in kinesiology* (5th ed.). Human Kinetics.

Weir, J. P., & Vincent, W. J. (2021). *Statistics in kinesiology: Instructor guide* (5th ed.). Human Kinetics.

The jamovi project. (2026). *jamovi* (Version 2.7) [Computer software]. https://www.jamovi.org

Use of artificial intelligence: the author led the work on this material from the initial idea to the finished content, designing it, setting the rules for writing and language use, and checking every part. Claude Code (Anthropic), an AI-based assistant, was used to write the programs that generate the slides, figures and documents, to draft text, and to check the language against the author's rules. The author is responsible for the content.